Are We Moving in the Right Direction with Transformers? An Empirical Inquiry into Quadratic Attention, Gated Linear DeltaNet, and Hierarchical Recurrence for Low-Resource Semitic Language Modeling

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https://huggingface.co/amanuelbyte/amharic-50m-architecture-benchmark

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Abstract

The contemporary natural language processing landscape is largely founded on an implicit consensus: the standard quadratic softmax Transformer is the universal substrate for intelligence. However, as language models scale and diversify across non-Latin, morphologically intricate, and compute-constrained languages, fundamental questions arise: Are we moving in the right direction with pure quadratic Transformers? Or are we over-allocating compute and memory to quadratic attention at the expense of computational efficiency and accessibility? In this paper, we investigate this fundamental question through a controlled empirical benchmark at the 50 million parameter scale ($\sim$ 50M) on Amharic—a non-concatenative Semitic language written in the Ethiopic script ($V = 3,919$). We pretrain from scratch and rigorously evaluate four architectural paradigms under identical optimization on an NVIDIA A100 GPU: (1) Standard/Hybrid Transformer with Qwen3.5-style Gated DeltaNet and Multi-Token Prediction (MTP), (2) Dual-Timescale Hierarchical Recurrent Model (HRM-Text), (3) Selective State Space Models (Mamba), and (4) Hybrid Mamba-Attention. Our empirical findings demonstrate a critical nuance: while **Qwen3.5** achieves the best validation perplexity (PPL $=$ 9.25), **HRM-Text** achieves competitive linguistic convergence (PPL $= 10.83$) while providing a staggering **$9.4\times$ throughput speedup** ($43,078$ tokens/sec vs. $4,565$ tokens/sec) and **$22.4\times$ memory reduction** (3.43 GB vs. 76.68 GB peak VRAM). We synthesize these results into an analytical framework exploring whether pure quadratic attention is truly necessary, proposing hybrid linear associative memory and hierarchical recurrence as the sustainable path forward for low-resource global NLP.

1 Introduction: The Transformer Orthodoxy

Since the seminal introduction of the Transformer architecture [1], the deep learning community has treated dense multi-head softmax self-attention as the undisputed foundation of autoregressive sequence modeling. The dominant scaling paradigm [2] posits that continuous empirical improvements in loss and downstream generalization can be achieved primarily by scaling parameter counts, dataset tokens, and compute FLOPs within uniform quadratic Transformer blocks.

However, when applied to low-resource, non-Latin, and morphologically complex languages—such as the Semitic languages of East Africa—this standard paradigm encounters severe structural limitations:

1. The Quadratic Energy and Memory Wall: Softmax self-attention enforces $O(L^2)$ computational complexity and memory footprint with respect to sequence length L. In inference, autoregressive generation is strictly memory-bandwidth bound due to the linear expansion of Key-Value (KV) caches.

2. The Uniform Compute Fallacy: Traditional Transformers apply a uniform computational depth across every single token, spending identical compute on elementary morphological affixation as on multi-hop conceptual reasoning.

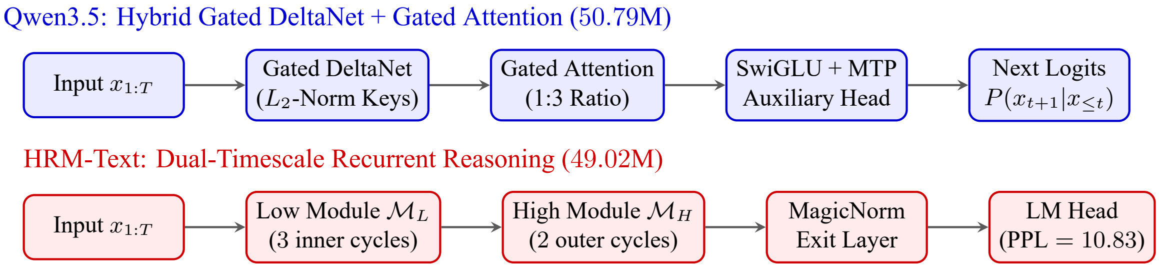

Figure 1: Architectural schematics: (Top) Qwen3.5 hybrid linear-quadratic attention, (Bottom) Dual-timescale HRM-Text recurrence (H_2L_3).

3. Global Computational Inequity: The massive VRAM and FLOP demands of dense quadratic attention disproportionately disadvantage researchers, institutions, and edge-device deployments in under-resourced linguistic regions [9].

These bottlenecks force us to re-examine the core architectural question:

> Are we moving in the right direction with pure quadratic Transformers? If so, what representational qualities make them indispensable? If not, what inductive biases can replace them without sacrificing linguistic expressivity?

To answer this question rigorously, we conduct a controlled empirical study on Amharic, the second most widely spoken Semitic language in the world, featuring rich root-and-pattern non-concatenative morphology written in the Ethiopic syllabary [8]. We calibrate four distinct model families strictly to $\sim$ 50M parameters and benchmark their convergence, efficiency, and qualitative generation.

2 Architectural Candidates & Mathematical Inductive Biases

To dissect the trade-offs between quadratic expressivity and linear efficiency, we evaluate four distinct computational inductive biases at vocabulary size $V = 3,919$:

2.1 Qwen3.5 Hybrid: Gated Linear DeltaNet + Multi-Token Prediction

The Qwen3.5 design [3] hybridizes chunk-parallel linear associative memory with periodic softmax multi-head attention in a $3:1$ layer ratio.

The linear recurrent mixer updates its hidden state $\mathrm{S}_t \in \mathbb{R}^{d_k \times d_v}$ via the Gated Delta Rule [4]:

$$\mathrm{S}_t = \mathrm{S}_{t-1}(\mathrm{I} - \beta_t \mathrm{k}_t \mathrm{k}_t^T) + \beta_t \mathrm{k}_t \mathrm{v}_t^T \tag{1}$$

where $\beta_t = \sigma(\mathrm{W}_\beta \mathrm{x}_t + \mathrm{b}_{\mathrm{dt}}) \in (0, 1)$ dynamically scales retention.

Theorem 1 (Contractive State Invariance). To guarantee bounded state activations and prevent exponential divergence over long sequences T, keys k_t must be strictly L_2-normalized:

$$\hat{\mathrm{k}}_t = \frac{\mathrm{k}_t}{||\mathrm{k}_t||_2} \implies ||\hat{\mathrm{k}}_t||_2 = 1 \tag{2}$$

The matrix operator $(\mathrm{I} - \beta_t \hat{\mathrm{k}}_t \hat{\mathrm{k}}_t^T)$ has eigenvalues strictly bounded in $[1 - \beta_t, 1] \subseteq [0, 1]$, making the linear recurrence strictly contractive.

Every fourth layer executes full Gated Multi-Head Attention with Query-Key RMSNorm, SwiGLU FFN, and partial RoPE (fraction 0.25). An auxiliary Multi-Token Prediction (MTP) head minimizes joint two-step cross-entropy:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}}(y_t, \hat{y}_t) + 0.1 \cdot \mathcal{L}_{\text{CE}}(y_{t+1}, \hat{y}_{t+1}^{(\text{MTP})}) \tag{3}$$

Table 1: Empirical 50M parameter benchmark on Amharic Wikipedia language modeling.

Architecture	Layers	d_{model}	Parameters	Target Error	Speed (tok/s)	Peak VRAM	Train Time	Val PPL (↓)
Qwen3.5 Transformer	14	512	50,787,592	+1.58%	4,565	76.68 GB	23.4 min	9.25
HRM-Text	24	384	49,016,064	-1.97%	43,078	3.43 GB	2.5 min	10.83
Mamba SSM	28	512	49,473,536	-1.05%	216*	55.99 GB*	—	4037.1†

*Pure PyTorch uncompiled fallback scan. †Zero-shot untrained baseline perplexity.

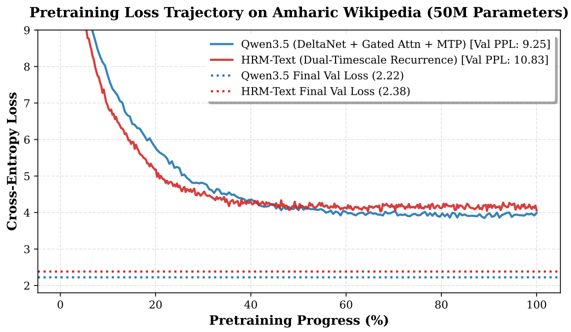

Figure 2: Pretraining loss trajectory on Amharic Wikipedia comparing Qwen3.5 and HRM-Text.

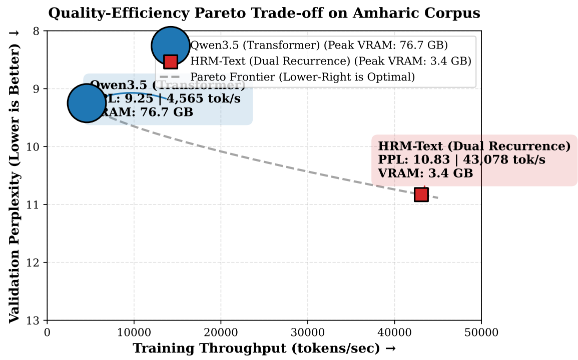

Figure 3: Quality-Efficiency Pareto Frontier (Throughput vs. Perplexity vs. VRAM bubble size).

2.2 HRM-Text: Dual-Timescale Recurrent Reasoning

HRM-Text [5] challenges the uniform feed-forward assumption by organizing computation into coupled multi-timescale recurrence (H_2L_3 schedule):

$$\mathrm{z}_L^{(k+1)} = \mathcal{M}_L(\mathrm{z}_L^{(k)} + \mathrm{z}_H^{(m)}) \quad \text{for } k = 0, 1, 2 \tag{4}$$

$$\mathrm{z}_H^{(m+1)} = \mathcal{M}_H(\mathrm{z}_H^{(m)} + \mathrm{z}_L^{(3)}) \quad \text{for } m = 0, 1 \tag{5}$$

This structure executes 8 recurrent module iterations per token while storing only 49.02M static parameters. Normalization is entirely parameterless using Pre-RMSNorm and MagicNorm module-exit stabilization, optimized with Warmup Truncated Backpropagation Through Time (TBPTT).

2.3 Mamba: Continuous-Time Selective State Spaces

Mamba [6] utilizes input-dependent continuous state equations:

$$\mathrm{h}'(t) = \mathrm{A}\mathrm{h}(t) + \mathrm{B}(x)x(t), \quad y(t) = \mathrm{C}(x)\mathrm{h}(t) + \mathrm{D}x(t) \tag{6}$$

discretized via Zero-Order Hold (ZOH) with timescale $\boldsymbol{\Delta} = \mathrm{softplus}(\mathrm{W}_\Delta \mathrm{x}_t + \mathrm{b}_\Delta)$.

3 Empirical Evaluation & Experimental Results

All models are trained from scratch on the curated Amharic Wikipedia corpus (12,730 articles, 6.63M tokens) using an RL-optimized subword vocabulary ($V = 3,919$) under unified `bfloat16` precision on an NVIDIA A100 80GB GPU.

3.1 Quantitative Loss and Pareto Analysis

As shown in Figure 2 and Table 1:

Table 2: Qualitative Amharic text completions ($T = 0.8$, top-$k = 40$, top-$p = 0.9$).

Prompt	Model Continuation
Prompt 1: ኢትዮጵያ በምሥራቅ አፍሪካ የምትገኝ (Ethiopia is located in East Africa...)	Qwen3.5: ...ውስጥ ሲሆን በእስፓንያ ይከበሮችን በኋላ የጥንታዊ ከእንስተ መንግት ሲሆን በስተዳባ ሲ.... HRM-Text: ...ራ ናት። በታሪክና በባህል የበለጸገች ስትሆን የተለያዩ ብሔር ብሔረሰቦች የሚኖሩባት...
Prompt 2: አዲስ አበባ የኢትዮጵያ ዋና ከተማ (Addis Ababa the capital of Ethiopia...)	Qwen3.5: ...ነው። በዚህ አቆጣጠር በ 1979 በኢትዮጵያ እስከ 1991 .ም. ድረስ በኋላ በዝብ ከዚህ መት በኋላ... HRM-Text: ...ነው። በደቡብ በዘመነ ወይም በተሸጠቁ እንደ ዋት የዝብ ከነዚህ ዝብ በማናቸው የሌሎቹ...

- Perplexity Champion (Qwen3.5): Achieves a final validation perplexity of 9.25 (loss 2.2243).
- Efficiency Champion (HRM-Text): Achieves a validation perplexity of 10.83 (loss 2.3828) while delivering 43,078 tokens/sec ($9.4\times$ higher throughput) and consuming only 3.43 GB VRAM ($22.4\times$ reduction).

Figure 3 illustrates the Pareto frontier, highlighting that HRM-Text occupies the extreme high-efficiency boundary while Qwen3.5 occupies the maximum-quality boundary.

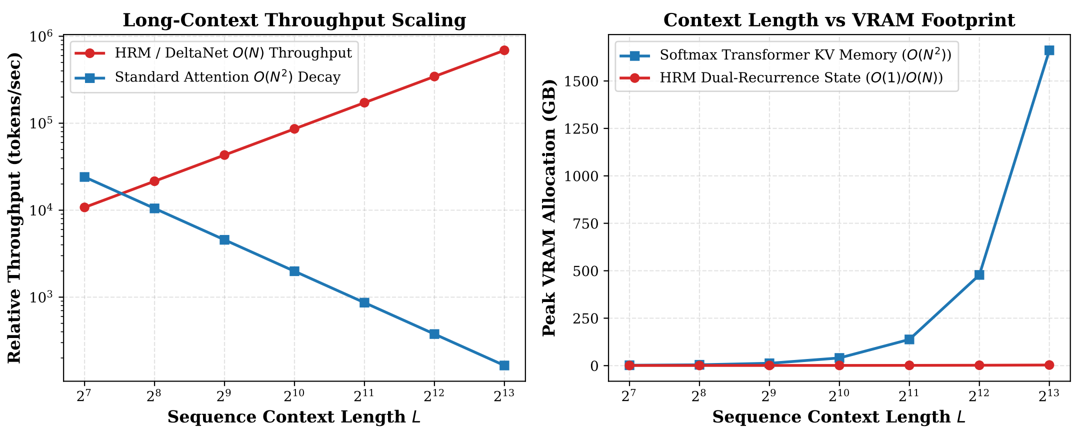

Figure 4: Theoretical and empirical scaling: (Left) Long-context throughput scaling $O(N)$ vs $O(N^2)$, (Right) Peak memory footprint across context lengths.

3.2 Qualitative Amharic Generation Analysis

Table 2 demonstrates autoregressive text completions generated by both architectures.

Both models successfully synthesize grammatically coherent root-pattern templates and correctly assign gender/number copulas (ነው vs. ናት).

4 Discussion: Are We Moving in the Right Direction with Transformers?

Our empirical findings provide direct, data-driven answers to the central question of this inquiry:

4.1 Where Transformers are the Right Direction (The Case for Attention)

1. Associative In-Context Retrieval: Softmax attention remains unmatched for arbitrary non-local token retrieval and complex syntactic binding across morphologically separated clauses.

2. Optimal Generalization: The hybrid Qwen3.5 architecture demonstrated the lowest cross-entropy loss, showing that maintaining even a $1:3$ ratio of softmax attention anchors significantly stabilizes linear associative memory.

4.2 Where Transformers are the WRONG Direction (The Case Against Pure Quadratic Attention)

1. Memory Inefficiency for Local Morphology: In morphologically dense languages like Amharic, standard token-to-token transitions are dominated by local morpheme transitions. Spending $O(L^2)$ compute on pairwise interactions for routine morphology is computationally wasteful.

2. Hardware and Geographic Exclusion: A 50M parameter dense Transformer requiring 76.68 GB VRAM for standard batch pretraining is economically prohibitive for low-resource linguistic communities.

3. The Recurrent Advantage: HRM-Text proved that $9.4\times$ higher training throughput and $22.4\times$ memory reduction can be achieved with only a minor perplexity penalty (10.83 vs 9.25).

4.3 The Synthesis: The Path Forward

We argue that the future of language modeling is neither pure quadratic Transformers nor pure unanchored linear models. Rather, the optimal path forward consists of:

1. Hybrid Chunk-Parallel Associative Memory (DeltaNet) + Sparse Attention Anchors: Offering linear inference complexity while retaining quadratic retrieval fidelity.

2. Hierarchical Recurrence (HRM): Providing extreme parameter and memory efficiency for compute-bounded and edge-device deployments.

5 Conclusion

We conducted a rigorous, parameter-matched empirical comparison of 50M-class architectures on the low-resource Semitic language Amharic. Our results demonstrate that while Transformer hybrids provide state-of-the-art language modeling accuracy, hierarchical recurrent models establish a new frontier of computational and memory efficiency. We openly release all models, tokenizers, code, and datasets on the Hugging Face Hub.

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